

Jo Gervantamir

Ottawa, ON

bsnlkhm@gmail.com — (613)-978-9207 — LinkedIn — GitHub — Portfolio



Portfolio

Education

Computer Programming Diploma (Co-Op)

Sept 2023 – Apr 2025

Algonquin College, Ottawa, ON Awards:

- Dean's Honour List (GPA 3.9)
- 1st Place, Hack the Hill 2023 — RSS feed generator with event-driven architecture (TypeScript, Svelte, Fastify)

Technical Skills

- **Languages:** Java, Python, C++, JavaScript/TypeScript, SQL, Bash, PHP, Dart
- **Libraries / Tools:** Spring Boot, Maven, Gradle, Tomcat, Payara, Hibernate/JPA, Sequelize, Swagger, Postman, DBaver, Eclipse IDE, ActiveMQ, Nagios, Wireshark, Packet Tracer
- **Web/Mobile Development:** React, Angular, Node.js, Express, NestJS, Flask, RESTful APIs, JAX-RS, Flutter, Dart
- **Databases:** PostgreSQL, MySQL, MSSQL, Oracle SQL, SQLite, MongoDB, Neo4j
- **Data & ML:** OCR (Tesseract, MICR E-13B), NLP, OpenCV, Web Scraping, Image Preprocessing Pipelines
- **DevOps & Security:** Docker, Kubernetes, Git/GitHub, Bitbucket, SonarQube, AWS, Linux/UNIX, Windows, VMware, OAuth 2.0, JWT, Keycloak, SSL/TLS
- **Other:** Agile/Scrum, SDLC, Software Requirements & Design Analysis, UML, SOLID, Figma

Relevant Experience

Full Stack Developer (Part-Time)

Dec 2025 – Jul 2026

CashmereCasa, Ottawa, ON

- Built and shipped a full-stack inventory + POS platform (**React 19**/TypeScript, Express, Sequelize, MySQL), replacing Excel workflows with transaction-sourced stock reconciliation (stock = receipts – sales) across **3K+ SKUs**, eliminating quantity drift.
- Built a **250+ endpoint** REST API across **43 route modules** over **56 Sequelize models**, with **Supabase** authentication (JWT verification via jose) and role-based access, plus a full audit- and deletion-log trail on every data mutation.
- Engineered a staff-facing **POS sales register** with barcode scanning, composite SKU generation (category/style/size/color), per-item discounts (percentage, fixed, gift/exchange), tax application, and real-time inventory auto-decrement on sale confirmation.
- Synced three storefronts — **Square POS**, **WooCommerce**, and **Shopify** — into one source of truth, collapsing **4** separate inventory systems. The Shopify integration spans **12 modules**: OAuth install flow, product/order/inventory sync, signature-verified webhooks, order deduplication, and a candidate-matching pass that lined a 3K-SKU catalog up against a live store without creating duplicates; scheduled cron jobs and per-integration sync logs catch whatever the webhooks miss.
- Built a bookkeeping module — CSV bank-statement import wizard, transaction categorization by expense type, a reconciliation matching engine, and P&L / HST / cashflow / expense reports.
- Delivered a vendor finance pipeline (invoice tracking, payment allocation, vendor valuation) and shipment landed-cost analysis (customs, freight, per-unit) across **50+ international shipments in 4 currencies**.
- Engineered a bulk CSV/Excel importer (ExcelJS) with confidence-scored column matching and dataset-wide typo correction, ingesting **20K+ rows** at **99%** auto-mapping accuracy.
- Launched a custom report builder with Excel/PDF export, barcode label printing, and a **color-blind-safe (Okabe-Ito)** UI for accessible chart and label legibility, cutting month-end reporting effort by **35 hrs**.
- Containerized the stack via a multi-stage **Docker** build and Docker Compose (MySQL + MinIO object storage + Express) with idempotent schema migrations, enabling weekly prod releases without DB downtime; documented **GCP** Compute Engine deployment.
- Built a **Playwright** E2E suite (**30 specs** covering auth, POS, inventory, accounting, and integrations) plus **67** server-side Jest test files, wired into **GitHub Actions** CI on every push and PR.

Full Stack Developer

Sep 2024 – Nov 2025

Ideabytes Inc. (Neology – ETBOS Highway Tollpoint System), Ottawa, ON

- Engineered distributed **Spring Boot** microservices (Traffic Sensor, Highway Map with custom SVG/Leaflet overlays) processing real-time sensor data across **4 major highways** and **60+ toll points** with multi-lane configurations.

- Architected event-driven pipeline on **ActiveMQ** — topic-based pub-sub for downstream fan-out and queue-based point-to-point for command processing — decoupling **10+ services** within a 50-service production ecosystem.
- Spearheaded enterprise SSO rollout via **Keycloak** + Active Directory (OAuth 2.0, RBAC), securing **200+ internal users** across **4 enterprise apps**.
- Delivered Flutter mobile PoC with payment tokenization and Nagios health-visualization API, accelerating stakeholder sign-off by **2 weeks**.
- Slashed pre-deployment defects by **80%** by establishing Bitbucket CI/CD pipelines with **SonarQube** quality gates, automated PR checklists, and SOLID-driven code reviews.
- Diagnosed and resolved **150+ production incidents** (~5/sprint) via HTTP traffic analysis, Spring Boot remote debugging, and Logback forensics.
- Drove requirements and design reviews with Product, translating business needs into UML specs that shipped **6 microservices across 27 Agile sprints**.

SW/ML Engineering Intern

May 2023 – Jul 2023

Ideabytes Inc. – Full Stack Internal Applications, Remote, Canada

- Engineered automated cheque-processing ML pipeline by training a custom **Tesseract OCR** model on MICR E-13B dot-matrix characters and banking transit symbols, eliminating **~20 hrs/week** of manual entry.
- Generated synthetic training corpus of **500+** images across varied print qualities and lighting using Python + **OpenCV**, expanding dataset coverage **3x**.
- Automated pipeline validation in Python to monitor model metrics and system health across training runs.
- Boosted character recognition accuracy from **67% → 91%** (+24 pts) through systematic experimentation with adaptive thresholding, deskewing, and noise reduction.

Organizer & Design Coordinator

May 2023 – Nov 2024

Hack the Hill — Tech Events and Hackathons, Ottawa, ON

- Designed **20+** Figma web layouts driven by user requirements gathering, achieving **92%** WCAG accessibility compliance.
- Secured **\$60K** in sponsorships and coordinated end-to-end logistics for events serving **600+** attendees.

Projects

Project Management Repository | Java, Hibernate, JavaScript, PostgreSQL

Aug 2025

- Built **25+** RESTful endpoints with Java EE (JAX-RS) and JPA/Hibernate, powering task management, auth, and reporting designed for multi-tenant use.
- Implemented JWT authentication with refresh-token rotation and RBAC across Admin, Project Manager, and Client roles, hardening access control on **50+ protected routes**.
- Optimized PostgreSQL schema with composite indexes on task queries, accelerating dashboard loads from **3.2s → 0.8s** (75% faster) for users with 1000+ tasks.
- Achieved **85%** code coverage through unit + integration suites (JUnit, Maven Surefire), catching **12+** regressions pre-release.

Medical Education School Management | Java EE, JPA, Postman, Maven

Jul 2025

- Delivered **76** RESTful endpoints spanning 8 resource domains and managing 1000+ records in production.
- Attained **100%** code coverage via JUnit + Maven Surefire continuous-test automation on Linux.

Transferable & Soft Skills

- Analytical problem solver with strong SDLC and requirements-analysis instincts
- Effective communication and cross-functional collaboration with Product, QA, and Design
- Resourceful, self-directed learner — continuously builds AI/ML, cloud, and cybersecurity skills via hackathons, LeetCode, and conferences
- Time management, attention to detail, and adaptability across Agile/Scrum teams
- Languages: English (Fluent), Mongolian (Native), German (Intermediate), French (A2 — Basic)